

MIT805 Big Data Project Report

FHVHV NYC Taxi Trip Dataset

1. Dataset Provenance and Description

The dataset consists of For-Hire Vehicle High Volume (FHVHV) trip records stored in 24 Parquet files. The notebook indicates data was sourced as monthly trip files covering January 2024 to December 2025. These files were available publicly from NYC Taxi & Limousine Commission. The dataset contains trip-level ride information including pickup and drop-off timestamps, trip distance, trip duration, fares, tips, driver pay, tolls, and location identifiers within New York. The dataset is appropriate for academic use because it contains operational transportation data rather than personally identifiable passenger information.

2. Dataset Size and Characteristics

Source: FHVHV trip data. Format: Apache Parquet. Number of files: 24. Rows: 483.06 million records. Compressed size: 10.93 GB. Uncompressed size: 13.41 GB. Date range: 01 January 2024 to 31 December 2025. Schema: 24 attributes including temporal, geographic and financial variables. Working dataset after cleaning contained approximately 417.96 million records.

3. Data Quality and Preparation

Missing-value analysis revealed substantial null values in `originating_base_num` and `on_scene_datetime`. These columns were removed. Duplicate checks identified 354 duplicate records which were removed. Invalid trips with non-positive duration (21,911 records) were excluded. Additional consistency checks compared `trip_time` against calculated pickup-to-dropoff duration, removing 37,617 inconsistent records. Fare values less than or equal to zero were removed. Outliers in `trip_time` and `trip_miles` were handled using the Interquartile Range (IQR) method. Potential concerns include uneven geographic representation, temporal bias across weekdays, missing operational fields, and the risk of misleading correlations between fare, distance and driver compensation.

4. Exploratory Data Analysis (EDA)

EDA showed strong temporal patterns. Saturday recorded the highest trip volume (70.1 million trips) while Monday had the lowest (51.6 million trips) as seen in Figure A2. Monthly and yearly trend analysis was conducted using bar charts as seen in Figure A1 and Figure A3. The number of trips increased slightly in 2025. The number of trips increased during the last 3 months of the year. A Heatmap (Figure A5) illustrated demand distribution by hour and weekday, showing significant variation across time periods. From Monday to Friday, most of the trips are requested around 8 am and around 5pm revealing passengers rely on these services to get to and from work. On Friday and Saturday evenings, it is evident that passengers are requesting many trips between 7PM and 12 PM. This reveals peak hours for requested trips. Correlation analysis (Figure A4) revealed positive

relationships between trip distance, trip duration, fares, tips and driver pay, indicating that longer trips generally produce higher revenue and compensation.

5. Analysis Using the 7 Vs of Big Data

Volume: The dataset demonstrated high-volume as two years of data constituted to 483 million records and over 10.9 GB compressed data justify distributed processing with Spark.

Velocity: Transportation trips are generated continuously as seen by the pick-up and drop-off times, requiring scalable ingestion and processing.

Variety: Dataset contains timestamps, categorical identifiers, geographic IDs, financial measures and operational flags.

Value: Supports transport planning, route optimization, demand forecasting and operational efficiency improvements.

Veracity: Data cleaning addressed missing values, duplicates, invalid durations and outliers to improve reliability.

Variability: Demand differs by weekday, month and hour, demonstrating changing usage patterns.

Visualization: Heatmaps, bar charts and correlation matrices were required to understand patterns hidden within hundreds of millions of records.

6. Business and Societal Value, Limitations

The dataset can support demand forecasting, fleet allocation, surge pricing strategies and urban transport planning. Insights may help reduce waiting times, improve operational efficiency and support evidence-based transport policy. Limitations include limited contextual variables such as weather, special events or demographic indicators, and potential bias caused by underrepresented locations or incomplete operational fields.

Key EDA Findings

- The analysis was conducted on a large-scale dataset comprising 483.06 million ride records across 24 Parquet files, covering the period January 2024 to December 2025.
- Following data cleaning, duplicate removal, validation checks, and outlier treatment, the final working dataset contained approximately 417.96 million records.
- The EDA revealed clear temporal patterns in ride demand, with Saturday recording the highest trip volume, indicating increased transportation activity during weekends.
- The dataset exhibited strong big-data characteristics, enabling the identification of demand trends, operational patterns, and relationships between trip duration, distance, fares, tips, and driver compensation.

Github link: <https://github.com/cara-jvr/mit-805-group-project.git>

References

New York City Taxi and Limousine Commission (TLC). (2026). *High Volume For-Hire Vehicle Trip Records (HVFHV) [January 2024-December 2025]*. City of New York. Retrieved from <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

Appendix A: Supporting Figures

Figure A1 *Distribution of trips by year.*

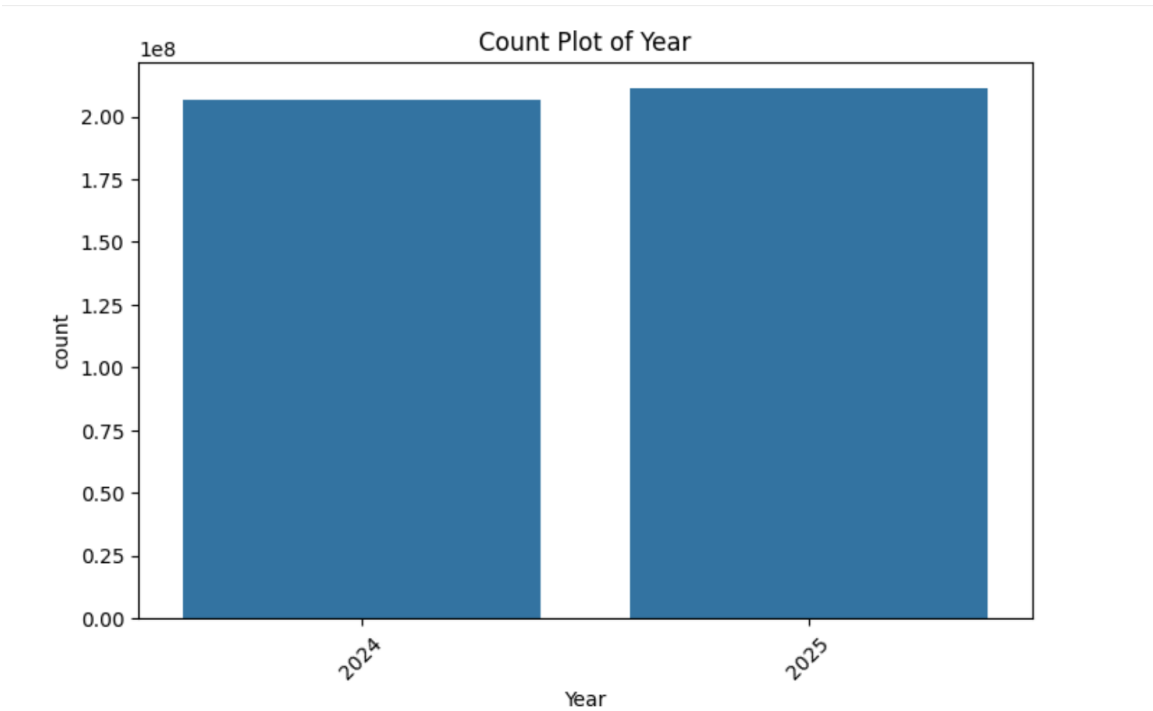


Figure A2 *Distribution of trips by weekday.*

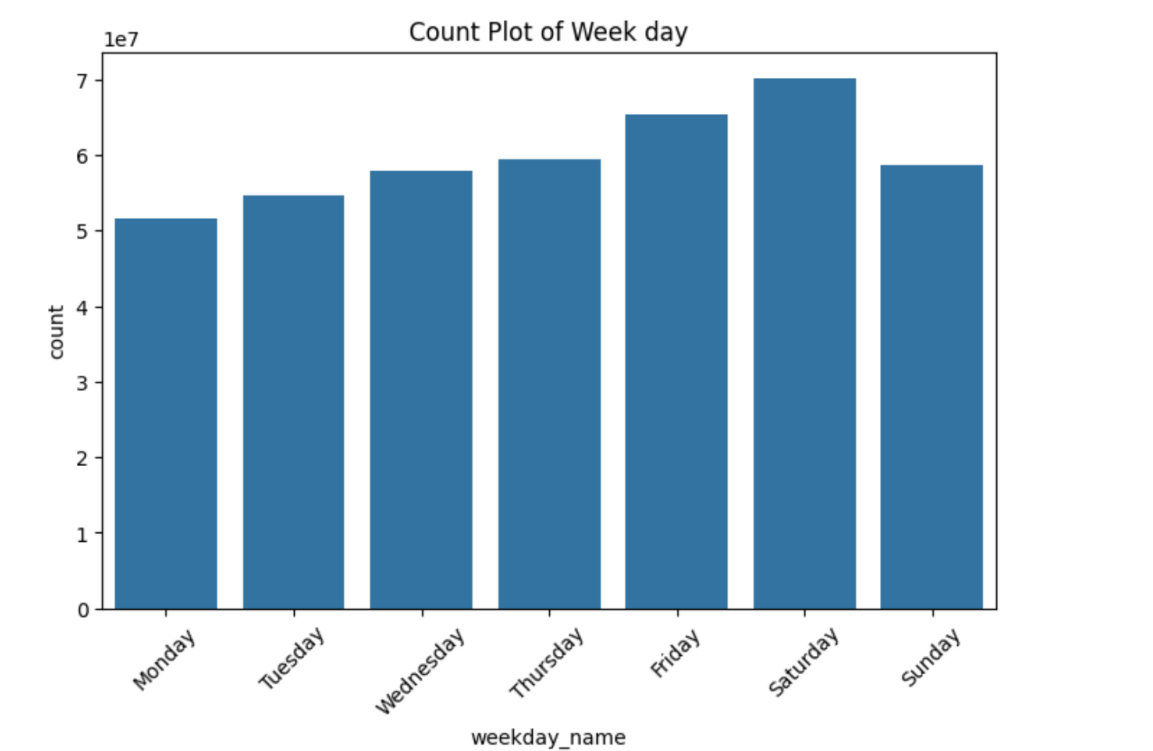


Figure A3 *Distribution of trips by Month*

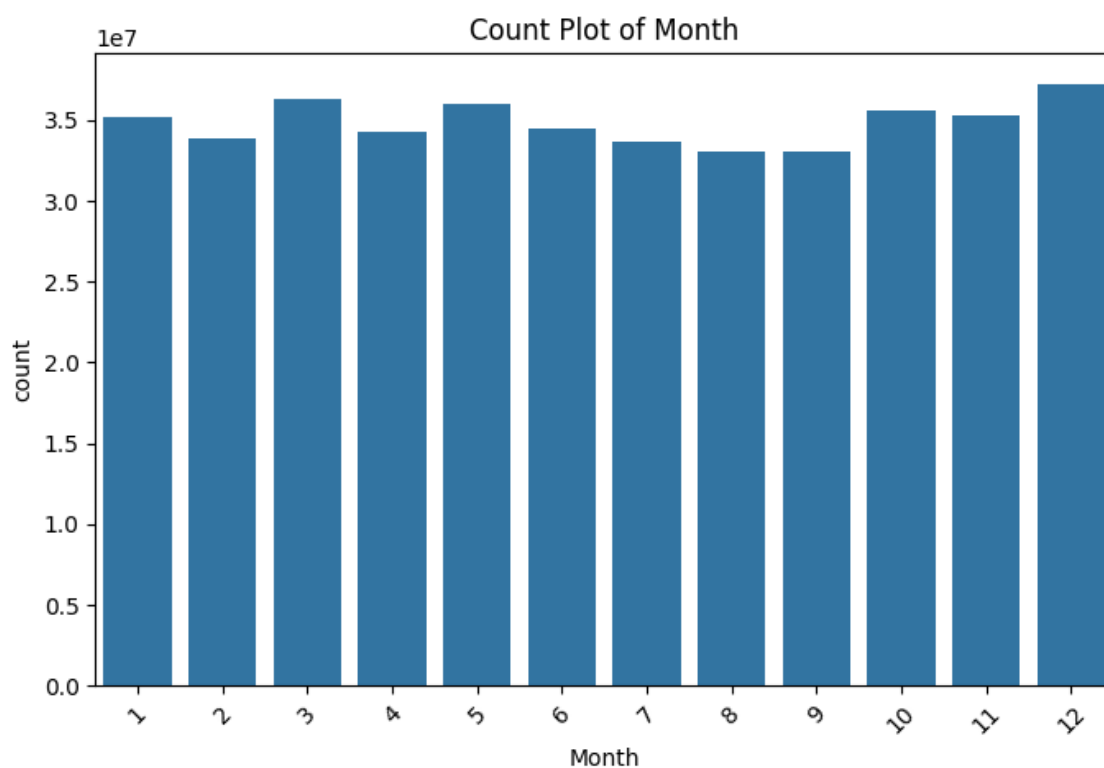


Figure A4 *Correlation heatmap of selected numerical variables.*

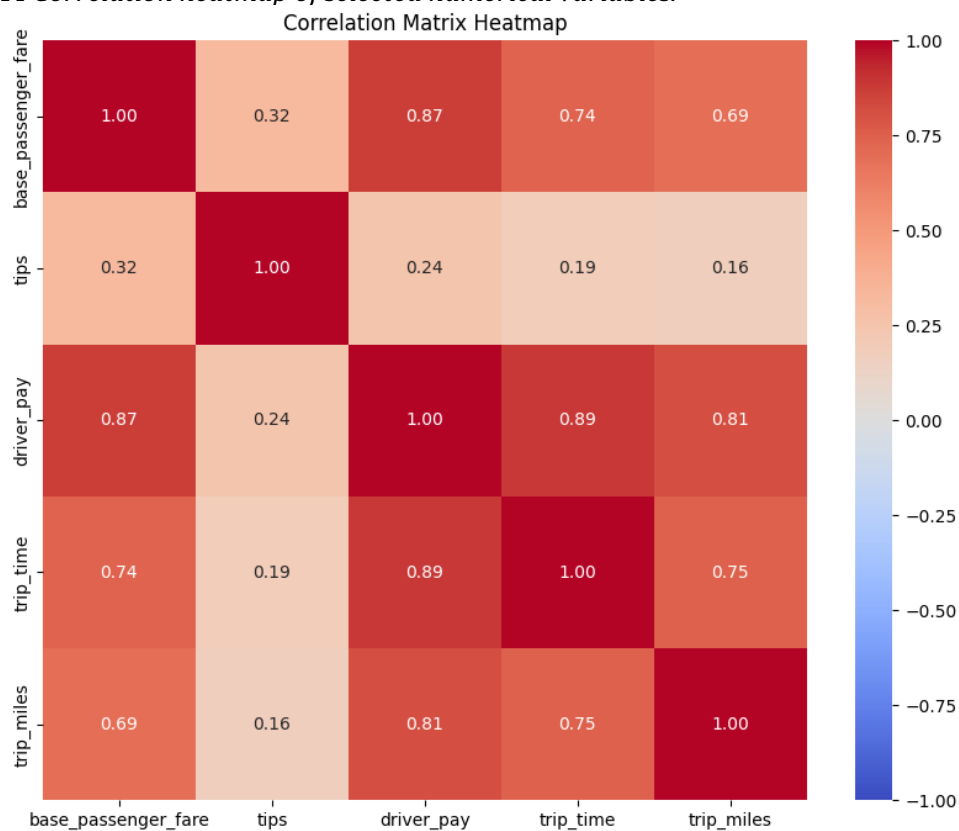


Figure A5 *Heatmap of day of week and hour of the day*

